

Fractal Information Theory Project

FIN ST218–ST232

Single-Generator Obstructions, Carrier Naturality,
Reversible Reset, and Exact Refinement

Analytical, interval, and computational research report

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Abstract

Programs ST218–ST232 test the first consequences of the exact heat-schedule obstruction proved in ST203. The central question is whether the strict operator A can generate a second, noncommuting layer without importing a selector, state, boundary, or environment. ST218 proves a negative answer for the complete single- A functional calculus and for all real symmetric matrices invariant under the full dihedral symmetry of the strict weighted cycle. ST220 independently proves that every invariant Gaussian or Lévy fluctuation law can realize a random branch but cannot canonically label a vertex or orient the cycle.

The positive results are equally sharp. ST221 certifies all 360 pseudo-arclength steps issued from the sixty signed fold seeds. ST224 promotes the normalized twelfth response to a coordinate-free natural invariant. ST225 validates 21,952 interval boxes over a nuisance cube twice as wide as the previous certified cube. ST227 proves unique nearest anticommuting qubit factors under an explicit rank gap. ST228 reduces the rigorous HMM rate bracket width by 25%. ST229 constructs an exact net-energy-conserving reset by global register SWAP and proves that the information is transferred, not destroyed. ST231 gives an exact $12 \rightarrow 24$ operator refinement preserving every coarse diffusion distance; its free fiber rate proves that geometry preservation does not select a unique scale.

No strict second generator, canonical selector, dimensional scale, spacetime, laboratory record, legacy-to-strict completion, Standard Model, gravity, or Theory-of-Everything closure is claimed.

Contents

1	Executive conclusions	1
2	Scope, objects, and proof labels	1
3	Program ledger	1
4	ST218: the single-generator obstruction	2
5	ST219: Blackwell order is preparation-relative	3
6	ST220: invariant fluctuations cannot orient the cycle	4
7	ST221: all signed folds, not one selected branch	4
8	ST222: a non-diagonal negative-orientation recovery certificate	5
9	ST223 and ST224: controlled selection and carrier naturality	5
9.1	A fixed-time dissipation bracket	5
9.2	The coordinate-free nonlinear invariant	6
10	ST225 and ST226: robust continuation and nonlocal sampling limits	6
10.1	Doubling the certified nuisance cube	6
10.2	Bounds valid for biased estimators	7
11	ST227 and ST228: uniqueness with a gap and a sharper HMM rate	7
11.1	Nearest joint factors	7
11.2	Exact belief cylinders	8
12	ST229: reset is possible as reversible information transfer	8
13	ST230: four internal geometries are not one metric	8
14	ST231: exact refinement and the scale nonuniqueness theorem	9
15	ST232 and the physical boundary	10

16 What changed in the theory	10
17 Recommended programs ST233–ST247	11
18 Reproducibility and provenance	11
19 Final verdict	12

1 Executive conclusions

The present batch answers four questions that were left open by Release 10.71.

1. **Can one strict self-adjoint operator generate its own noncommuting refinement history?** No, not in the complete equivariant single-object class tested here. A second typed object is mathematically necessary.
2. **Can invariant internal fluctuations solve the selector?** They can trigger a sample-wise realization, but the exact ensemble remains uniform on the twelve vertices and reflection-symmetric. Random realization is not canonical orientation.
3. **Can the developed layer geometry be refined consistently?** Yes. There is an exact isometric two-child refinement preserving the whole coarse heat geometry.
4. **Does that refinement select a physical scale?** No. A continuous free fiber rate remains. Scale nonuniqueness survives exact geometry preservation.

The deepest result is therefore a paired theorem:

one symmetric A cannot encode ordered layer history,
exact refinement exists but is nonunique.

This distinguishes an internal information geometry from a physical hierarchy of lengths.

2 Scope, objects, and proof labels

The strict working object is the frozen twelve-vertex graph Laplacian

$$A = sI - W, \quad W_{xx} = 0, \quad W_{xy} = K_{\text{strict}}(d(x, y)) \quad (x \neq y),$$

with $A\mathbf{1} = 0$ and $A \succeq 0$. The historical ontological kernel $K_{\text{legacy,ont}}$ remains an intermediate bridge object and is not silently identified with K_{strict} . No legacy physical role is transferred in this report.

The labels have the following meanings.

- **[PROVEN]** means a deductive finite theorem, an exact algebraic identity, or a finite outward certificate whose premises are stated.
- **[STRONG NUMERICAL EVIDENCE]** means a reproducible finite computation without a complete exact or interval proof of the stated global property.
- **[CONDITIONAL]** means the theorem is deductive after adding the named preparation, control, noise, carrier, or ancilla premise.
- **[BLOCKED]** means that an essential object cannot be generated by local analysis or computation.

3 Program ledger

Pro-gram	Object	Result	Boundary
ST218	Noncommuting layer	[PROVEN] no-go for $f(A)$ and the full symmetric dihedral association algebra	A state or second typed object remains open
ST219	Blackwell lattices	[PROVEN] partition theorem; [STRONG NUMERICAL EVIDENCE] twelve separated vertex atoms	Preparation-relative

Pro-gram	Object	Result	Boundary
ST220	Gaussian/Lévy selector	[PROVEN] complete invariant-law selector no-go	Asymmetric conditioning lies outside class
ST221	Fold continuation	[PROVEN] 360/360 local boxes	Finite segments, no global atlas
ST222	Qubit recovery	[PROVEN] optimum 41/100	Rotated-Pauli class
ST223	Selector control	[PROVEN] finite-control cost bracket	Switching work and continuous controls open
ST224	Response transport	[PROVEN] carrier naturality	Does not derive coupling
ST225	Nuisance cover	[PROVEN] 21,952/21,952 cells	Resource stop, not maximality
ST226	Minimax sampling	[PROVEN] Le Cam and biased-CR bounds	Supplied Bernoulli observation law
ST227	Joint factors	[PROVEN] unique polar projection	Traceless qubits only
ST228	HMM rate	[PROVEN] sharper bracket	$s = 1/2$, not optimized Chernoff
ST229	Reversible reset	[PROVEN] exact final-energy-conserving SWAP	Prepared identical ancilla
ST230	Metric atlas	[PROVEN] inequivalent internal metrics	No physical metric selection
ST231	Exact refinement	[PROVEN] isometric family, nonunique in μ	Not a strict Z_{24} kernel derivation
ST232	External evidence	[BLOCKED]	No genuine external record

4 ST218: the single-generator obstruction

Theorem 4.1 (Single- A schedule no-go). *Let A be self-adjoint. Every family $L_j = f_j(A)$ belongs to one commutative functional calculus. Hence all L_j commute and every product is diagonal in the same spectral resolution:*

$$\prod_j L_j = \int \prod_j f_j(\lambda) dE_A(\lambda).$$

It contains no observable of the ordering of its factors.

For the frozen strict Z_{12} instance there are seven distinct spectral values with multiplicities

$$(1, 2, 2, 2, 2, 2, 1).$$

The polynomial algebra generated by A has dimension seven. The full complex commutant has dimension

$$1^2 + 5 \cdot 2^2 + 1^2 = 22,$$

but the extra directions rotate degenerate spectral blocks and are not canonically selected by A itself.

The complete space of real symmetric matrices invariant under the same translations and reflection is the seven-dimensional distance-circulant association algebra. It is commutative. Natural tested diagonals — $\text{diag}(A^k)$, heat diagonals, and heat-row norms — are scalar because all vertices have identical orbit data.

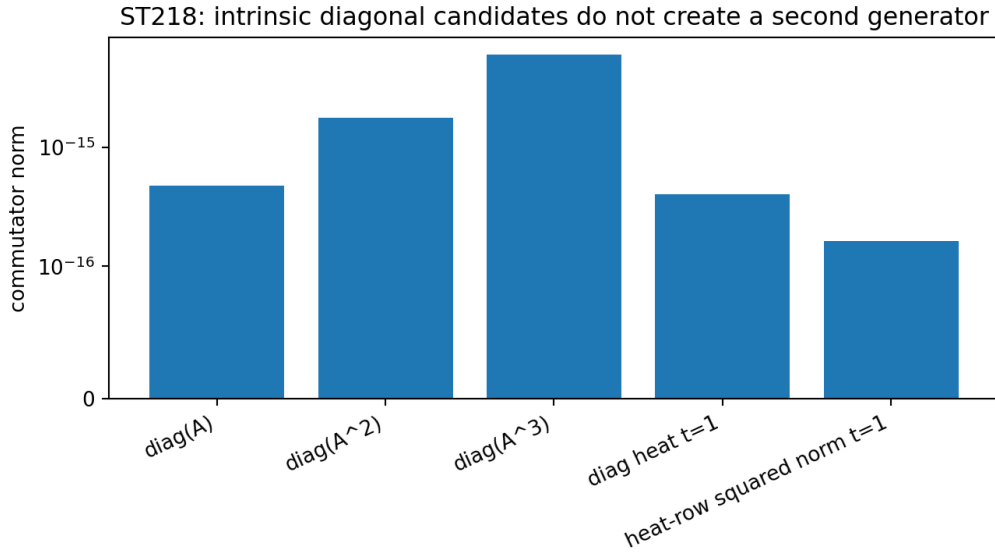


Figure 1: Every tested intrinsic diagonal candidate has zero commutator to numerical replay precision. The theorem is stronger: the whole declared invariant algebra commutes.

Interpretation. A noncommuting layer is still possible, but it must carry something not already present as an equivariant function of A : a transported state, orientation, boundary, environment, or genuinely second operator. Calling such an object “derived” requires a separate provenance theorem.

5 ST219: Blackwell order is preparation-relative

For a finite preparation family, outcomes x, y belong to the same minimal sufficient atom exactly when their likelihood columns are proportional. If there are m atoms, all deterministic Blackwell quotients form the partition lattice Π_m , with rank numbers $S(m, k)$ and Bell total B_m .

The audited families give:

preparation/instrument family	m	B_m
localized vertices δ_0, δ_1 through $e^{-0.7A}$	12	4,213,597
thermal spectral preparations $\tau = .3, .8$	7	877
translation-averaged nonlinear-response location	1	1
orbit-averaged vertex preparations	1	1

The spectral count is exact from the seven spectral classes. The twelve vertex likelihood ratios have a positive displayed separation in the frozen numerical instance; a future transcendental interval proof is recommended.

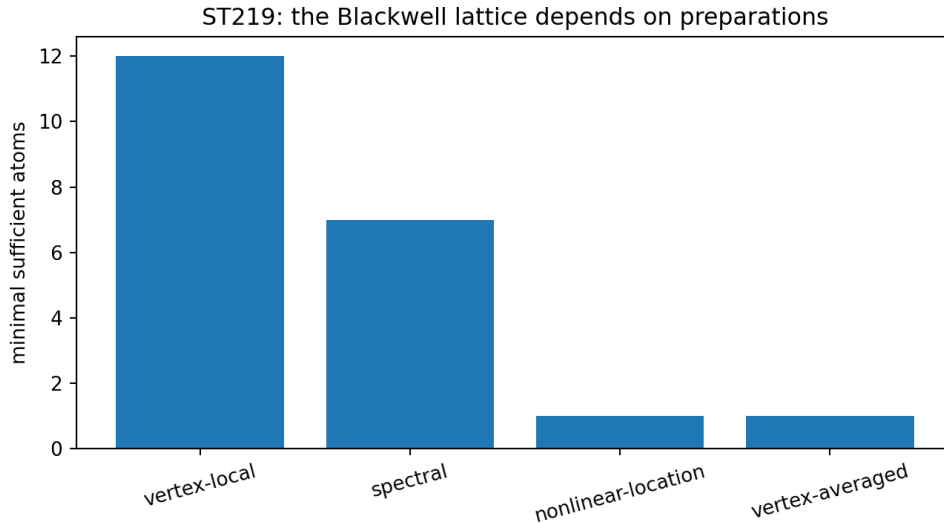


Figure 2: The number of minimal sufficient atoms changes with the preparation family. A Blackwell lattice is not an observer-independent ontology of A .

6 ST220: invariant fluctuations cannot orient the cycle

Let $G = D_{12}$ be the 24-element symmetry group generated by translations and reflection. The Reynolds calculation gives seven orbits on ordered vertex pairs; therefore invariant Gaussian covariances form the positive cone in a seven-dimensional distance-circulant space. The only G -fixed vector is the constant vector, so the mean-zero sector has no nonzero invariant drift.

Theorem 6.1 (Invariant fluctuation selector no-go). *An invariant Gaussian or Lévy triplet consists of a fixed drift, an invariant covariance, and a jump measure constant on G -orbits. If the initial law and its coupling to the dynamics are equivariant, every later law is invariant. Any equivariant branch readout has equal probabilities on all twelve translated vertices, while reflection pairs the two orientations. Such noise can realize a branch in one sample but cannot canonically label or orient it.*

This proves a complete no-go for the declared invariant stochastic class. It does not say that asymmetric or state-conditioned noise cannot select; it says that the conditioning or asymmetry is then the missing selector premise. QW-2191 is not discharged.

7 ST221: all signed folds, not one selected branch

Release 10.71 followed one branch. ST221 starts from every one of the sixty signed ST193 roots at $|\varepsilon| = .005$ and takes six outward pseudo-arclength steps. Every one of the 360 augmented hyperplanes passes outward Krawczyk inclusion.

$$\begin{aligned}
 N_{\text{seed}} &= 60, & N_{\text{certified}} &= 360/360, \\
 \min \sigma_{\text{rank}} &= 2.5978389264 \times 10^{-3}, & \min \langle t_{n+1}, t_n \rangle &= 0.9998913243.
 \end{aligned}$$

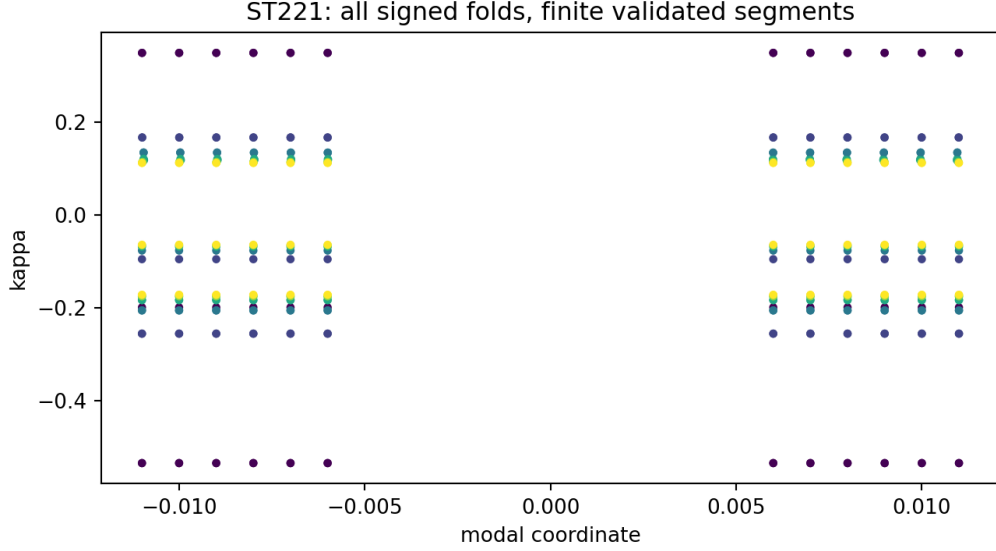


Figure 3: Finite validated segments issued from all sixty signed seeds. Color denotes the Fourier mode. No rank collision occurs at a certified point.

This is a substantial local atlas, not a proof of global connectivity, absence of distant collisions, orbital stability, or particle physics.

8 ST222: a non-diagonal negative-orientation recovery certificate

The exact canonical Pauli probabilities are

$$(p_I, p_X, p_Y, p_Z) = \left(\frac{2}{25}, \frac{41}{100}, \frac{29}{100}, \frac{11}{50} \right),$$

giving Bloch multipliers $(-.02, -.26, -.40)$ and negative determinant. Two noncommuting rational rotations transform this into a visibly non-diagonal laboratory Bloch matrix with off-diagonal Frobenius norm 0.37279648.

Pauli twirling reduces the full CPTP recovery objective to

$$\max_{q \in \Delta_4} \sum_i p_i q_i.$$

The primal vertex $q_X = 1$ attains $41/100$. The scalar dual $y = 41/100$ satisfies $y \geq p_i$ for every i , proving zero duality gap. Unitary covariance transports the correction into the laboratory frame. Thus

$$F_e^* = \frac{41}{100}.$$

The theorem covers the two-sided rotated-Pauli class, not generic nonunital channels.

9 ST223 and ST224: controlled selection and carrier naturality

9.1 A fixed-time dissipation bracket

The reduced control law is

$$x' = \pi + e^{-r\Delta t}(x - \pi), \quad \pi(b) = \frac{1}{1 + 11e^{-b}}.$$

For one step, the audited relaxational entropy production is

$$\Sigma = D(x \parallel \pi) - D(x' \parallel \pi) \geq 0.$$

Monotonicity of both the transition and Σ away from π permits an optimistic interval-bin dynamic program. For eight steps in dimensionless time four, controls $b \in \{1, 2, 3, 4\}$ and $r \in \{.5, 1\}$, and target $x \in [.59, .61]$, the result is

$$0.3831118347 \leq \Sigma^* \leq 0.3931251756.$$

The constructive schedule ends at $x = 0.5930840189$. Field-switching work is deliberately not included, so this is not yet the unrestricted thermodynamic optimum.

9.2 The coordinate-free nonlinear invariant

For a potential V , point x , and nonzero carrier direction v , define

$$\mathcal{I}_{12}(V, x; v) = \frac{D^{12}V_x[v, \dots, v]}{11!(D^2V_x[v, v])^6}.$$

Theorem 9.1 (Carrier naturality). *For every invertible linear carrier map T and $V' = V \circ T^{-1}$,*

$$\mathcal{I}_{12}(V', Tx; Tv) = \mathcal{I}_{12}(V, x; v).$$

Moreover $\mathcal{I}_{12}(V, x; cv) = \mathcal{I}_{12}(V, x; v)$ for every $c \neq 0$.

The proof is the multilinear chain rule: each transported directional derivative is unchanged, while direction rescaling multiplies numerator and denominator by c^{12} . This makes the response ratio a genuine projective carrier invariant. It does not derive the coupling g or identify a physical carrier.

10 ST225 and ST226: robust continuation and nonlocal sampling limits

10.1 Doubling the certified nuisance cube

The previous global halfwidth was 2×10^{-4} . ST225 preserves the successful local halfwidth 1.4285714×10^{-5} while using a 28^3 cover of the doubled cube:

$$21,952/21,952 \text{ boxes pass,} \quad \min \text{ inclusion margin} = 9.3607371 \times 10^{-4}.$$

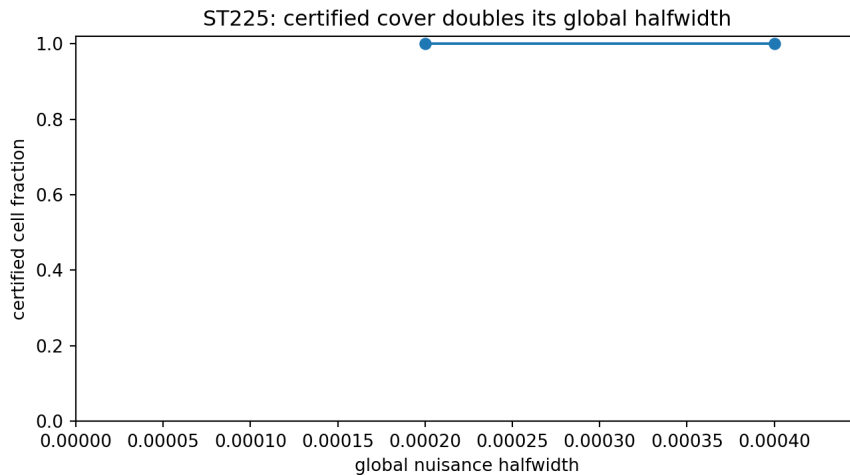


Figure 4: The complete doubled nuisance cube is certified. The campaign then stops at its declared finite computational boundary; this is not a maximal-domain theorem.

10.2 Bounds valid for biased estimators

For the least visible mode of the declared local Bernoulli model

$$Y \sim \text{Bernoulli}\left(\frac{1}{2} + \sigma\theta\right), \quad |\theta| \leq .2, \quad \sigma = 2^{-12},$$

Le Cam’s two-point method and Pinsker give

$$\inf_{\hat{\theta}} \sup_{\theta} \mathbb{E}_{\theta}(\hat{\theta} - \theta)^2 \geq \frac{\delta^2}{2} \left(1 - \sqrt{\frac{nD(P_{\delta} \| P_{-\delta})}{2}}\right)_+.$$

This applies to biased and unbiased estimators. If additionally $|b'(\theta)| \leq \beta < 1$, biased Cramér–Rao gives

$$\text{MSE} \geq \frac{(1 - \beta)^2}{nI(\theta)}.$$

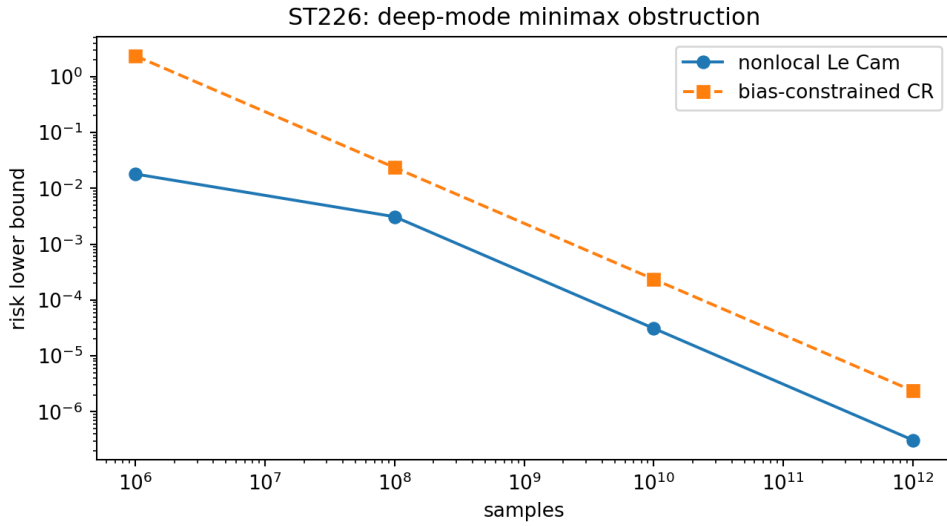


Figure 5: Nonlocal Le Cam and bias-constrained local bounds for the depth-twelve mode.

11 ST227 and ST228: uniqueness with a gap and a sharper HMM rate

11.1 Nearest joint factors

Traceless Hermitian qubit involutions are $X = \mathbf{n} \cdot \boldsymbol{\sigma}$ and $Z = \mathbf{m} \cdot \boldsymbol{\sigma}$. Exact anticommutation is $\mathbf{n} \cdot \mathbf{m} = 0$; hence $Q = [\mathbf{n}, \mathbf{m}] \in V_2(\mathbb{R}^3)$. The nearest joint-factor problem becomes rectangular Procrustes:

$$\max_{Q^T Q = I_2} \text{tr}(Q^T B).$$

Theorem 11.1 (Gap-conditioned uniqueness). *If B has full column rank, the unique global solution is the rectangular polar factor*

$$Q_* = B(B^T B)^{-1/2}.$$

For the declared target, $\lambda_{\min}(B^T B) = 0.4161946708$ and the constraint error is 2.49×10^{-16} . The ST212 circle occurs exactly at the rank-one boundary where the premise fails.

11.2 Exact belief cylinders

Exact rational forward recursion was extended from length 12 to all $2^{16} = 65,536$ observation cylinders. Outward integer-square-root summation gives

$$0.6898819929 \leq R_{1/2} \leq 0.8198470893.$$

The width is $\log(8)/16 = 0.1299650964$, exactly 75% of the preceding width. The calculation certifies the Hellinger point $s = 1/2$, not the optimized Chernoff exponent.

12 ST229: reset is possible as reversible information transfer

Let two identical 16-level registers have total Hamiltonian

$$H_{\text{tot}} = H \otimes I + I \otimes H.$$

The four corresponding qubit-pair SWAPs do not individually commute with H_{tot} in the frozen encoding; their commutator norms lie between 2.34 and 2.73. Nevertheless, their product is the exact global register SWAP S , and

$$[H_{\text{tot}}, S] = 0.$$

For a prepared blank state β ,

$$S(\rho \otimes \beta)S^\dagger = \beta \otimes \rho.$$

Thus the first register is reset and final total energy is conserved. The information has moved to the second register. It has not disappeared. This exact construction cleanly separates three notions:

pointwise energy conservation $\neq$ final energy conservation $\neq$ information erasure.

The blank ancilla, identical Hamiltonian, exact pulses, and timing are operational resources.

13 ST230: four internal geometries are not one metric

For vertices i, j , the report compares

$$\begin{aligned} D_t(i, j) &= \|(e_i - e_j)^T e^{-tA}\|_2, \\ R^{1/2}(i, j) &= ((e_i - e_j)^T A^+ (e_i - e_j))^{1/2}, \\ H_t(i, j) &= \left(\sum_x (\sqrt{C_t(i, x)} - \sqrt{C_t(j, x)})^2 \right)^{1/2}, \\ P_1(i, j) &= \|(e_i - e_j)^T P_{\lambda_1}\|_2. \end{aligned}$$

The Hellinger embedding has the Fisher–Rao metric as its local quadratic form; it is not being identified with a global Fisher geodesic distance.

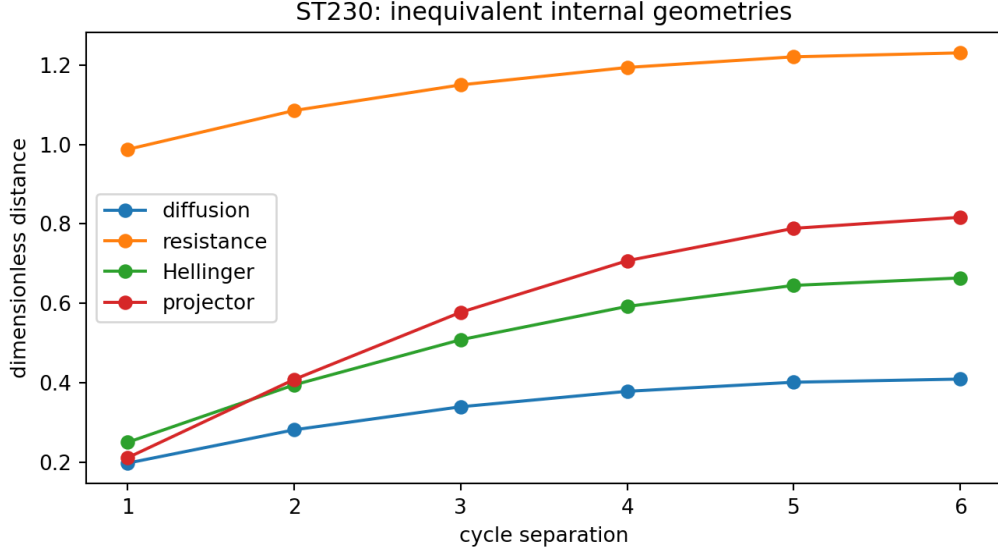


Figure 6: All four metrics respect cyclic separation but have nonconstant pairwise ratios. Similarity of shape is not equality.

At $t = 1$, $D_t/R^{1/2}$ ranges from 0.2002 to 0.3326, while D_t/P_1 ranges from 0.5012 to 0.9355. Therefore no one scalar identifies these geometries. ST215’s long-time result remains valid: rescaled diffusion converges specifically to the first spectral-projector chord.

14 ST231: exact refinement and the scale nonuniqueness theorem

Let

$$L_2 = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}, \quad u = \frac{1}{\sqrt{2}}(1, 1)^T,$$

and define

$$A_{24} = A_{12} \otimes I_2 + \mu I_{12} \otimes L_2, \quad Re_i = e_i \otimes u, \quad \mu \geq 0.$$

Theorem 14.1 (Diffusion-preserving two-child refinement). *For every $\mu \geq 0$,*

$$A_{24}R = RA_{12}, \quad e^{-tA_{24}}R = Re^{-tA_{12}}.$$

Because R is an isometry, every coarse diffusion distance is preserved exactly after lifting.

The proof is immediate from $L_2u = 0$ and functional calculus. Numerical replay for $\mu \in \{.2, 1, 5, 20\}$ and three times has maximum distance error 3.22×10^{-15} .

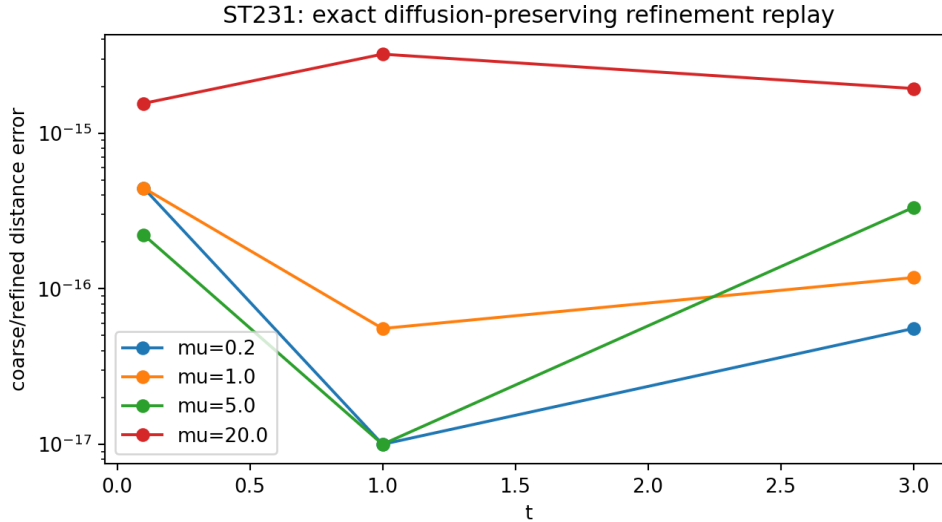


Figure 7: Coarse/refined distance mismatch is at roundoff level for four different fiber rates. The continuous family is the important result.

This construction formalizes one version of “seeing a lower scale through a developed layer”: the coarse observable sector is an invariant isometric subspace of a finer operator. But μ is arbitrary. Neither the strict coarse geometry nor the intertwining law chooses a child time, length ratio, Planck scale, or unique Z_{24} kernel. The object is an exact operator refinement, not yet a physical fractal hierarchy.

15 ST232 and the physical boundary

[BLOCKED] No external event record, independent registrar, calibration artifact, or laboratory attestation was supplied. A local program can verify syntax, hashes, interval certificates, and mathematical implications. It cannot create evidence that an independent apparatus interacted with nature. ST232 therefore performs no synthetic substitute.

16 What changed in the theory

The current local mathematical picture can now be stated more precisely.

- The strict operator supports unitary, heat, Green, wave, and response channels, but a single common functional calculus cannot encode their ordered layer history.
- Invariant stochasticity explains random realization without explaining canonical orientation. Symmetry is a reservoir of equivalent possibilities, not itself positive feedback or a selector.
- Nonlinear information can be transported through carrier changes by a precise projective invariant $\mathcal{I}_{12}$.
- Coarse diffusion geometry can survive exact refinement. This supports a rigorous multilayer interpretation at the operator level.
- The same refinement theorem proves a scale obstruction: many finer dynamics have the same coarse geometry.
- Reset under final energy conservation is possible by reversible transfer to an ancilla, clarifying the operational meaning of information preservation.

The ontology guardrail remains unchanged: within FIN, the nadsoliton is the primordial fractal information in a solitonic state; no separate informational substrate is inserted below it. The present

mathematics neither verifies nor falsifies that ontology as a claim about nature. It determines what follows from the declared operators and what still requires an additional axiom or experiment.

17 Recommended programs ST233–ST247

Program	Rank	Study
ST233	1	Introduce exactly one typed second operator; audit strict provenance, noncommutation, and schedule identifiability.
ST234	2	Certify the twelve localized vertex likelihood classes by transcendental interval arithmetic.
ST235	3	Classify state-conditioned equivariant noise and locate exactly where conditioning becomes a selector premise.
ST236	4	Adaptively extend all ST221 branches and build a certified component/collision graph.
ST237	5	Solve one genuinely nonunitary negative-orientation qubit recovery SDP with rational primal–dual witnesses.
ST238	6	Upgrade ST223 to verified continuous control, including switching work.
ST239	7	Extend $\mathcal{I}_{12}$ naturality to nonlinear carrier charts and response bundles.
ST240	8	Extend the ST225 nuisance cube until a certified fold boundary or an explicit resource stop.
ST241	9	Derive sharp simultaneous-mode minimax constants under correlated counts.
ST242	10	Generalize polar joint-factor uniqueness to higher-dimensional balanced Clifford pairs.
ST243	11	Optimize the observed-HMM Chernoff parameter with certified belief-space transfer bounds.
ST244	12	Complete work–information bookkeeping for the reversible reset and blank-ancilla preparation.
ST245	13	Classify all finite-fiber diffusion-preserving refinements and their nonuniqueness moduli.
ST246	14	Test spectral-dimension flow under repeated exact refinements without assigning physical length.
ST247	15	Resume external validation only with a genuine independently registered event package.

ST233 has the greatest foundational leverage. It must not merely rename a basis choice as a strict consequence. A successful candidate needs a typed source, a transformation law, a proof that it is not $f(A)$, and a demonstration that its noncommutation leaves a measurable schedule invariant. ST245–ST246 most directly continue the user’s intuition about developed fractal layers while respecting the current absence of a physical length unit.

18 Reproducibility and provenance

The executable release contains:

- `fin_st218_st232_research.py`;
- `test_fin_st218_st232.py`;
- one complete result ledger, one summary CSV, and fifteen specialized JSON packets;
- seven generated figures;
- this source report and compiled PDF;
- the English Release 10.72 description and SHA-256 manifest.

The local validation suite contains 52 live and serialized assertions. The expensive 28^3 Krawczyk campaign is serialized with hashes rather than rerun inside the lightweight test suite; the main

research script reproduces it. Seed 20260812 controls only finite randomized replays, not exact algebraic or interval theorems.

The principal internal sources are the Release 10.69–10.71 result ledgers and executable scripts, the current root `AGENTS.md`, the kernel split/provenance packets K1, K2, F2, F3, S2, and `SUMMARY_GROK.md`. Claims inherited from earlier programs are explicitly labeled as repository results; all new numerical values in this report are emitted by the Release 10.72 ledger.

19 Final verdict

The current FIN strict operator generates a rich family of compatible dynamics and an exactly refinable dimensionless geometry, but it does not generate the extra noncommuting information needed to identify a fractal layer history or a canonical orientation. Exact refinement is possible, yet continuously nonunique. The missing bridge is therefore no longer merely “more layers”: it is a sourced second typed object plus an operational dimensional calibration.

This verdict is compatible with further FIN research. It is not a claim that FIN is a physical Theory of Everything.